1. Gamma glutamyl transferase catalyses the reaction of glutathione (inside the cell) with the neutral amino acid (in the lumen) to form gamma glutamyl amino acid and cysteinyl glycine in the cytoplasm of the brush border membrane cell.
2. The glutamyl amino acid is then cleaved to give the free amino acid and oxo-proline inside the cytosol. The net result is the transfer of an amino acid across the membrane.
3. Oxo-proline is converted to glutamate. Cysteinyl glycine is cleaved into cysteine and glycine.
4. Glutathione is regenerated from the three amino acids.
5. The transport of one molecule of any amino acid along with regeneration of GSH requires 3 molecules of ATP.

After release into the cytosol, the amino acids are released into the blood stream and are transported to the liver by the entero-hepatic circulation.

Some oligopeptides, larger peptides or proteins can cross the brush border membrane by a process called transcytosis. This particularly happens in infants whereby the imunoglobulins in mother's milk can be transferred to the child. The enterocytes located in the Crypts of Lieberkuhn are generally involved in this process. However, this mechanism operates only to a lesser extent in adults.

Factors affecting absorption

- Glutathione is required for absorption of amino acids and its regeneration should be effective for good absorption.
- Amino acids transported by the same transporter compete with each other for absorption. Therefore, high concentrations of one particular amino acid will have an influence on the other.
- Salicylates decrease the intestinal absorption of amino acids, particularly that of tryptophan
- Dinitrophenol and indomethacin also inhibit the absorption of amino acids.

2.2.2.3 Digestion and absorption of lipids

The major difficulty in digestion of fats is that all the enzymes that digest fats are hydrophilic in nature, while fats are insoluble in water. This problem is circumvented by a process called emulsification. During emulsification, the fat is dispersed into smaller droplets, and their surface tension is reduced; and surface area is increased.

This process is facilitated by :

- the Detergent action of the Bile salts.
- Mechanical mixing i.e. chewing and peristalsis, where the surface area is increased.
- Presence of Phospholipids for micelle formation.

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